



**RAJALAKSHMI
ENGINEERING COLLEGE**

An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING**

CS23432 – Software Construction

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE

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Semester : IV
Academic Year : 2024-2028

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EXP NO: 1	STUDY OF AZURE
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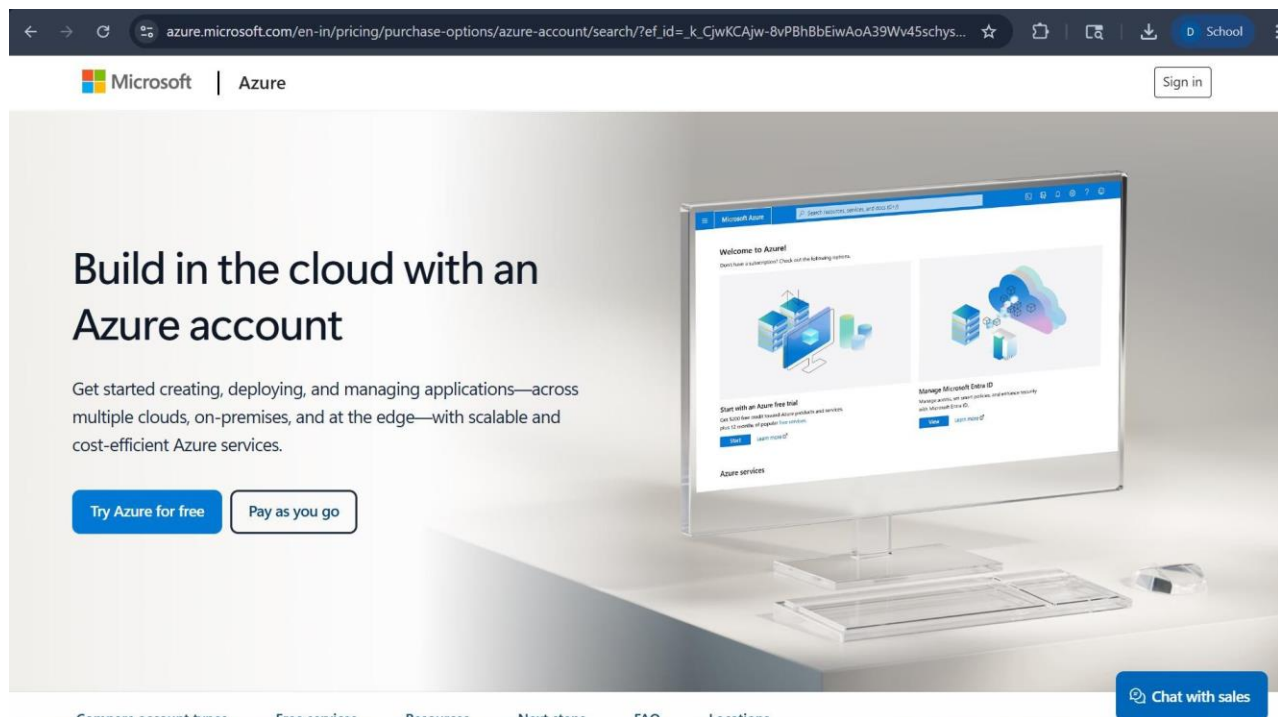
AIM:

To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

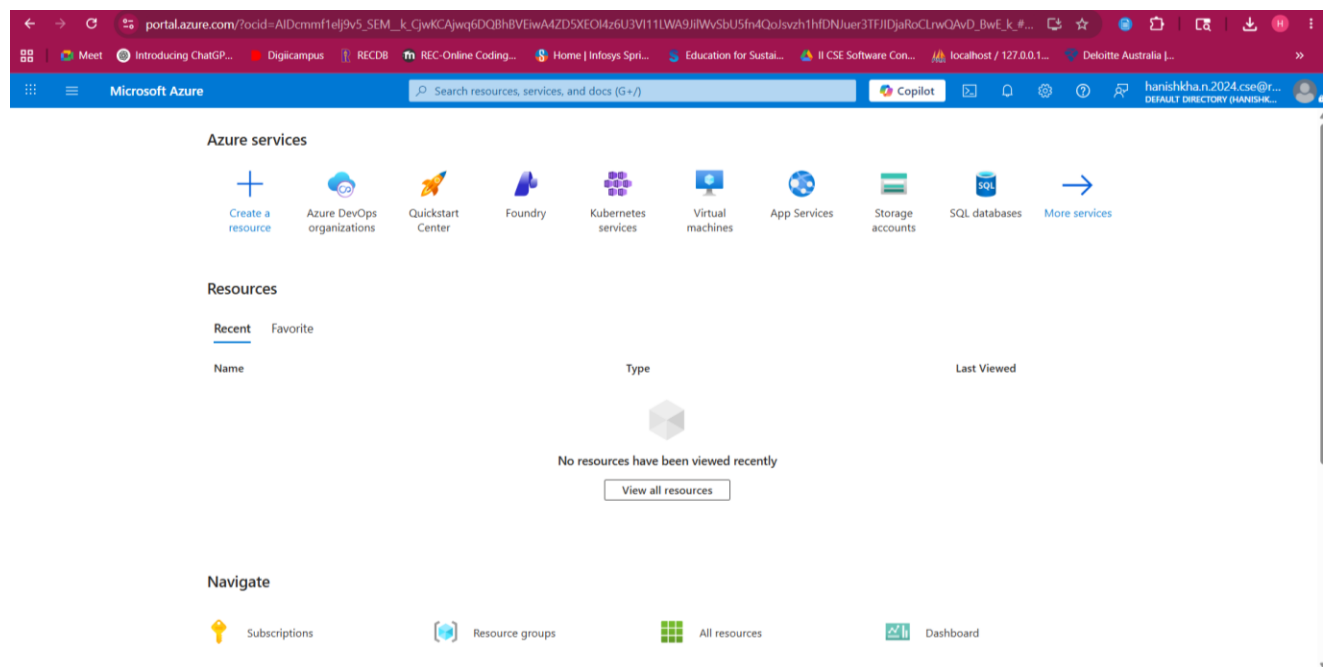
INSTALLATION

1. Open your web browser and go to the Azure website: <https://azure.microsoft.com/en-us/getstarted/azure-portal>. Sign in using your Microsoft account credentials.

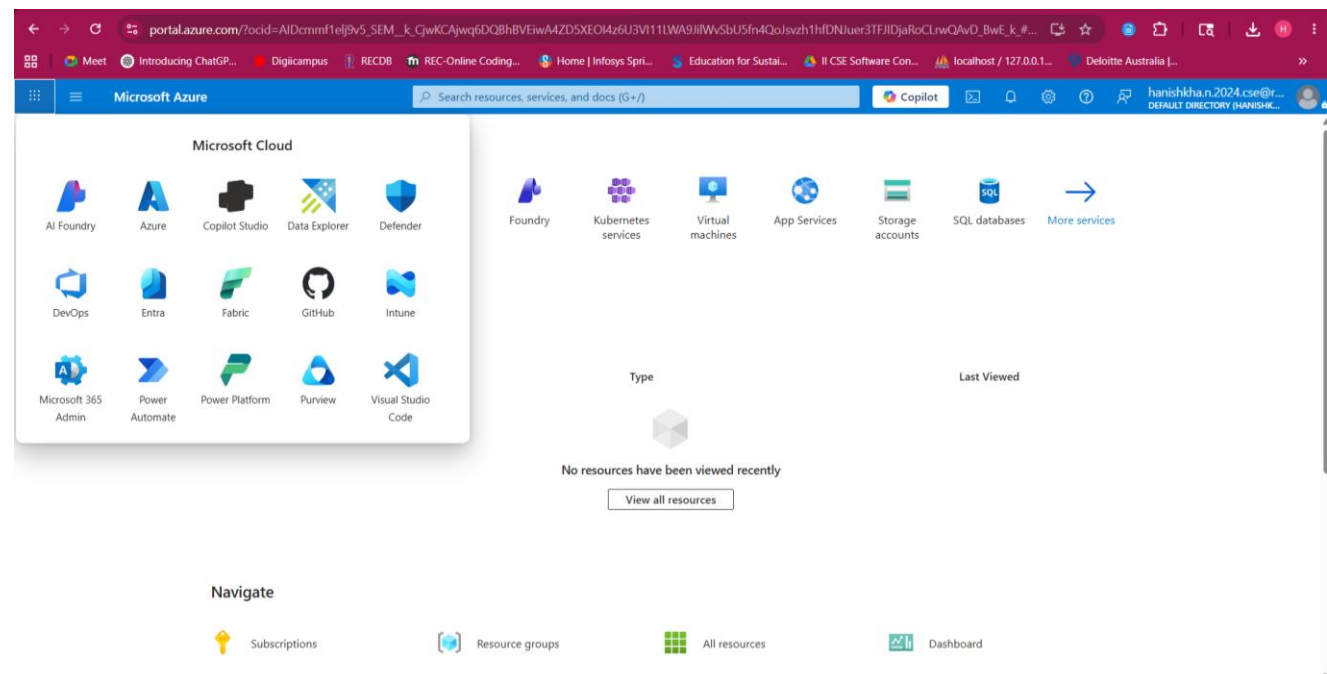
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



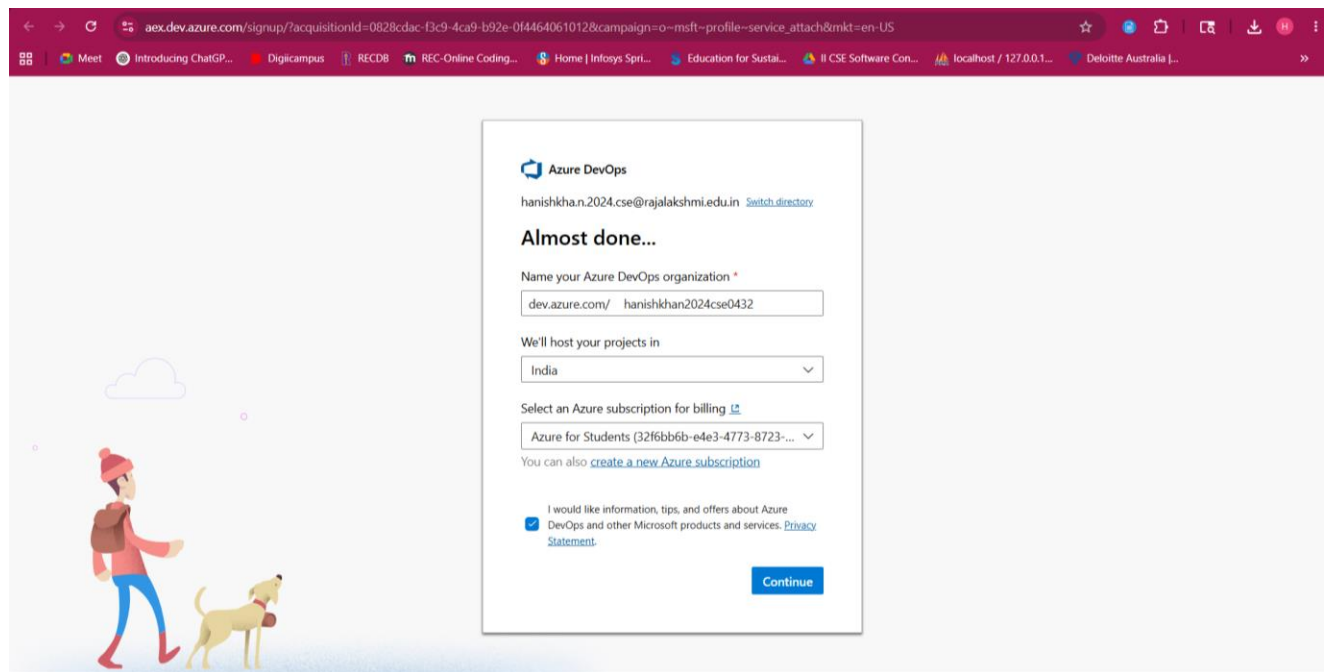
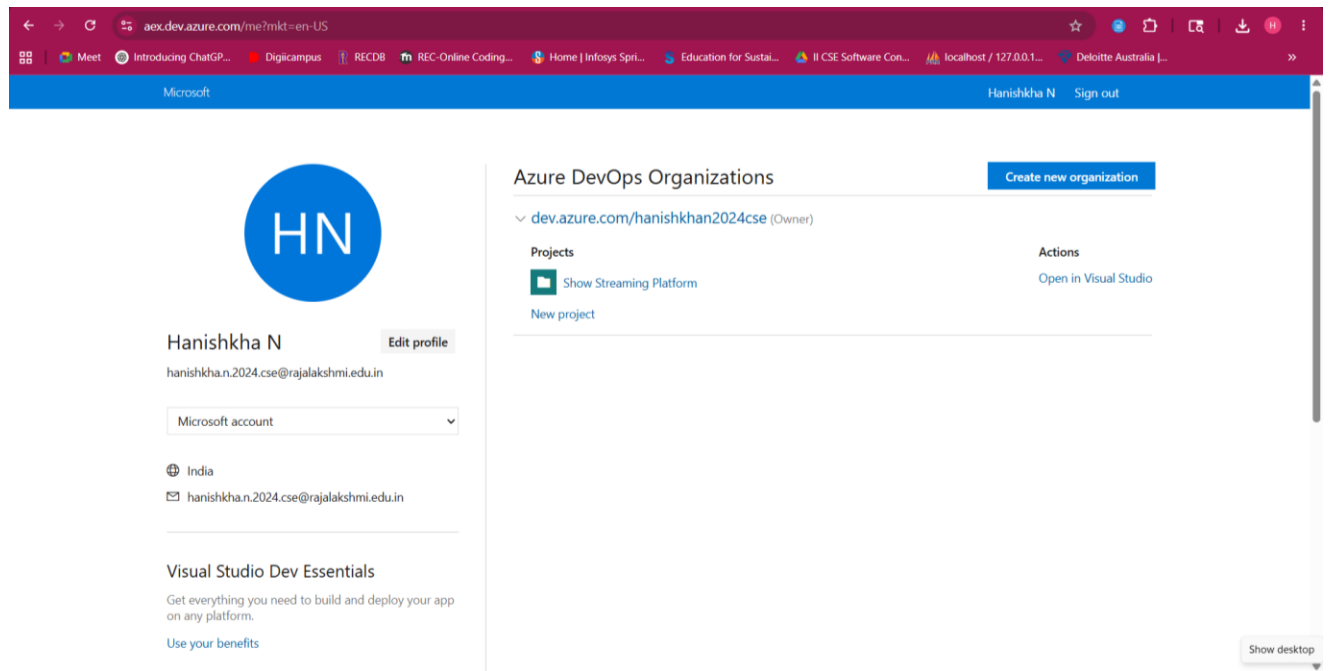
2. Azure home page



3. Open DevOps environment in the Azure platform by clicking 9 dots



4. Click on the DevOps icon and create an organization and you should be taken to the Azure DevOps Organization Home page.



RESULT :

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

EXP NO: 2	FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS
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AIM :

To list out the Functional and Non-Functional Requirements for our project (Shows Streaming Platform).

FUNCTIONAL REQUIREMENTS :

Functional requirements define the specific behaviors, services, and functions that a system must execute. They capture the intended behavior of the system, usually expressed as inputs, behavior, and outputs.

1. User Authentication:

- User registration (email & password)
- User login and logout
- Delete account
- Session handling

2. User Profile:

- Create and set up profile
- View and edit profile
- Profile picture
- User preferences (language, theme)

3. Home Page:

- Display trending shows
- Display popular and recommended shows
- Display recently added shows
- Organize shows by genre

4. Show Details Page:

- Show title and poster image
- Show description or summary
- Show genre and language
- Show release year and season details
- Show rating information

5. Search Functionality:

- Search shows by name
- Search shows by genre
- Search shows by language
- Search shows using cast or actor names
- Display relevant search results

6. Admin Management:

- Admin can add, update, or delete shows

- Admin manages content details

7. Favorites / Watchlist:

- Add shows to favorites
- View saved watchlist

NON FUNCTIONAL REQUIREMENTS :

Non-functional requirements specify the criteria used to judge the operation of a system rather than specific behaviors. They define the system's quality attributes, constraints, and performance characteristics

1. Performance:

- The system shall load pages quickly.
- The system shall provide fast search results.

2. Attractive UI Design:

- The system shall have an attractive user interface.
- The interface shall be user-friendly and easy to navigate.

3. Advanced Search Mechanism:

- The system shall support advanced search functionality.
- Search shall use show descriptions and cast details.

4. Multiple Language Support:

- The system shall support multiple languages.
- Users shall be able to switch languages easily.

5. Age Restrictions:

- The system shall enforce age restrictions for certain shows.
- Restricted content shall be accessible only to eligible users.

6. Data Security and Privacy:

- The system shall securely store user data.
- User information shall be protected from unauthorized access.

7. Customizable Themes:

- The system shall allow users to change themes.
- Light and dark modes shall be supported.

8. Reviews and Ratings:

- The system shall display reviews and ratings clearly.
- The system shall show average ratings for shows.

9. Scalability:

- The system shall support the addition of new shows.
- The system shall handle an increasing number of users.

10. Availability and Reliability:

- The system shall be available with minimal downtime.
- The system shall provide reliable access to information.

RESULT:

The Functional Requirements and Non-Functional Requirements for the project is listed successfully.

EXP NO: 3	PROBLEM STATEMENT
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AIM:

To prepare PROBLEM STATEMENT for your given project.

PROBLEM STATEMENT:

Most digital consumers face severe “choice paralysis” and fragmented viewing experiences when navigating vast content libraries across isolated platforms, spending more time scrolling through titles than actually enjoying entertainment. Building a centralized online shows streaming and content aggregation platform solves this critical industry bottleneck, allowing organizations to provide a highly personalized environment where users can seamlessly manage multiple individual sub-profiles, maintain highly customized wish lists for future viewing, and receive precise, filtered search results tailored to their specific tastes. Furthermore, this comprehensive tool directly helps viewers track their specific consumption progress, such as saving exact timestamps and current episodes in a long-running series, while simultaneously offering an interactive feedback loop through structured user reviews and community ratings designed to improve overall content discovery, algorithmic recommendations, and long-term audience engagement.

In addition to enhancing the front-end user experience, this centralized platform will serve as a powerful data-driven asset for content administrators and stakeholders looking to optimize their media delivery workflows. By integrating robust analytics engines behind the scenes, the system will securely monitor aggregate consumption trends, peak streaming hours, and popular search patterns without compromising individual user privacy. This rich pool of behavioral data will allow organizations to make informed, proactive decisions regarding content acquisition and licensing, ensuring that the library evolves in direct alignment with shifting audience demands. Ultimately, by bridging the gap between content overload and intuitive navigation, the platform transforms a traditionally overwhelming browsing process into a streamlined, highly engaging, and modern digital entertainment ecosystem

RESULT :

Thus ,the Problem Statement was stated successfully.

EXP NO:4	AGILE PLANNING
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AIM:

To prepare the AGILE PLANNING

THEORY:

Agile Planning is a part of the Agile methodology, which is a project management style with an incremental, iterative approach. Instead of using an in-depth plan from the start of the project—which is typically product related—leaves room for requirement changes throughout and relies on constant feedback from end users. With Agile planning, a project is broken down into smaller, more manageable tasks with the ultimate goal of having a defined image of a project's vision. Agile planning involves looking at different aspects of a project's tasks and how they'll be achieved, for example:

- Roadmaps to guide a product's release schedule
- Sprints to work on one specific group of tasks at a time
- A feedback plan to allow teams to stay flexible and easily adapt to change

User stories, or the tasks in a project, capture user requirements from the end user's perspective. Essentially, with Agile planning, a team would decide on a set of user stories to action at any given time, using them as a guide to implement new features or functionalities in a tool. Looking at tasks as user stories is a helpful way to imagine how a customer may use a feature and helps teams prioritize work and focus on delivering value first.

Steps in Agile planning process

1. Define vision
2. Set clear expectations on goals
3. Define and break down the product roadmap
4. Create tasks based on user stories
5. Populate product backlog
6. Plan iterations and estimate effort
7. Conduct daily stand-ups
8. Monitor and adapt

RESULT:

Thus the Agile plan was completed successfully.

EXP NO: 5

CREATING USER STORY AND EPIC

AIM:

Create User Story and Epic for the Shows Streaming Platform.

1. User Story :

The screenshot displays the Azure DevOps interface for a User Story. The left sidebar shows the navigation menu with 'Work items' selected. The main content area shows the details of a User Story titled '9 View Search History in Dashboard' created by Gayathri Jayam S. The story is in the 'New' state, located in the 'Show Streaming Platform' area, 'Iteration 1'. The description is: 'As a user, I want to view my previous searches in the dashboard so that I can quickly revisit shows or movies I searched for earlier.' The acceptance criteria are: 'System should save user search history automatically.', 'User should be able to view recently searched content in the dashboard.', 'Clicking a history item should redirect the user to the related content page.', and 'Search history should update whenever a new search is made.' The right sidebar shows the 'Planning' section with Story Points (2), Priority (2), and Risk (3 - Low). The 'Deployment' section has a note about tracking releases. The 'Development' section has an 'Add link' button.

Azure DevOps hanishkhan2024cse / Show Streaming Platform / Boards / Work items

Work Items Back to Work Items

USER STORY 9

9 View Search History in Dashboard

Gayathri Jayam S 0 Comments Add Tag

Save Follow

Updated by Gayathri Jayam S: Just now

Details

Description

As a user, I want to view my previous searches in the dashboard so that I can quickly revisit shows or movies I searched for earlier.

Acceptance Criteria

- System should save user search history automatically.
- User should be able to view recently searched content in the dashboard.
- Clicking a history item should redirect the user to the related content page.
- Search history should update whenever a new search is made.

Planning

Story Points
2

Priority
2

Risk
3 - Low

Classification

Value area
Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

2.Epic :

The screenshot displays the Azure DevOps web interface for a project named 'Show Streaming Platform'. The left sidebar contains navigation options: Overview, Boards, Work items, Boards, Backlogs, Sprints, Queries, Delivery Plans, Pipelines, Artifacts, and Project settings. The main content area shows an Epic titled '2 Home Page and Content Discovery' by user 'Hanishkha N'. The Epic is in the 'New' state and is associated with the 'Show Streaming Platform' area and iteration. The description of the Epic is: 'The **Home Page & Content Discovery** epic focuses on providing users with a personalized and engaging browsing experience. It enables users to easily discover shows through curated categories such as trending, popular, recommended, recently added, and genre-based collections. The goal is to help users quickly find content they are interested in without needing to search explicitly, improving engagement and retention through intelligent recommendations and intuitive UI design.' The right sidebar contains a 'Planning' section with fields for Priority (3), Risk (2 - Medium), Effort, Business Value, Time Criticality, and Start Date. Below this is a 'Deployment' section with a note: 'To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)'. The bottom section is labeled 'Development'.

RESULT:

Successfully created and managed Epics and User Stories for the Show Streaming Platform project in Azure Devops

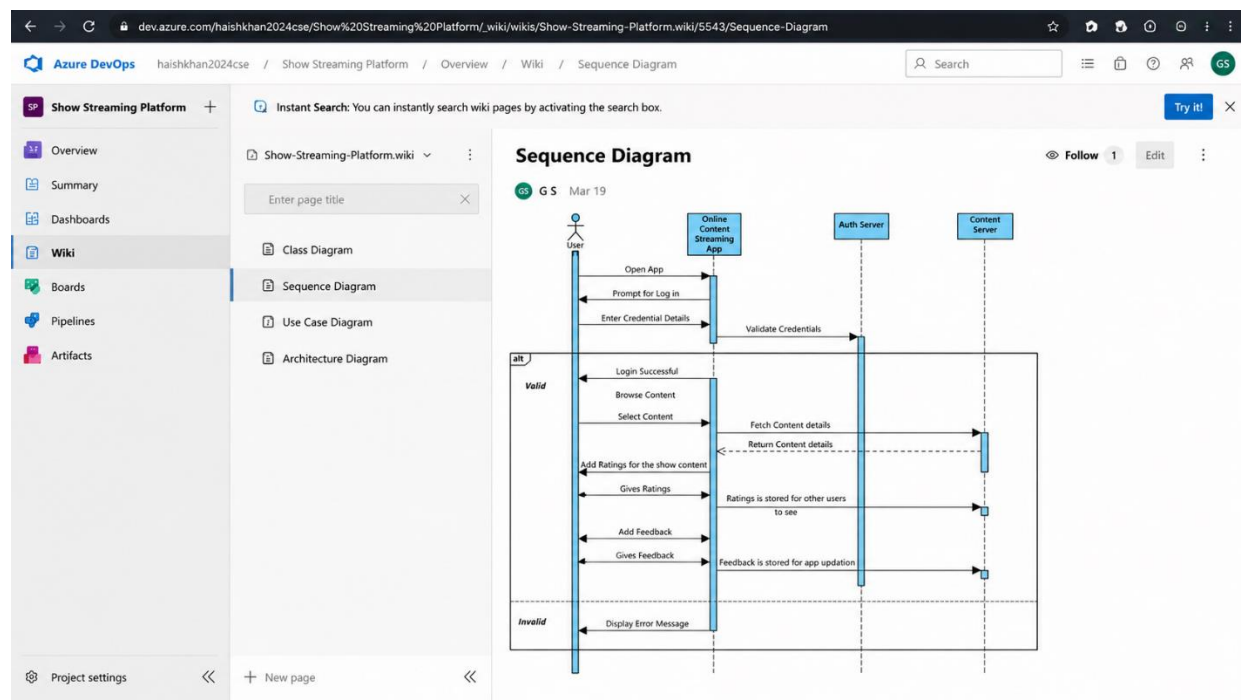
EXP NO: 6

SEQUENCE DIAGRAM

AIM:

To Design a Sequence Diagram for the given Project.

6. SEQUENCE DIAGRAM



RESULT:

The Sequence Diagram is designed successfully.

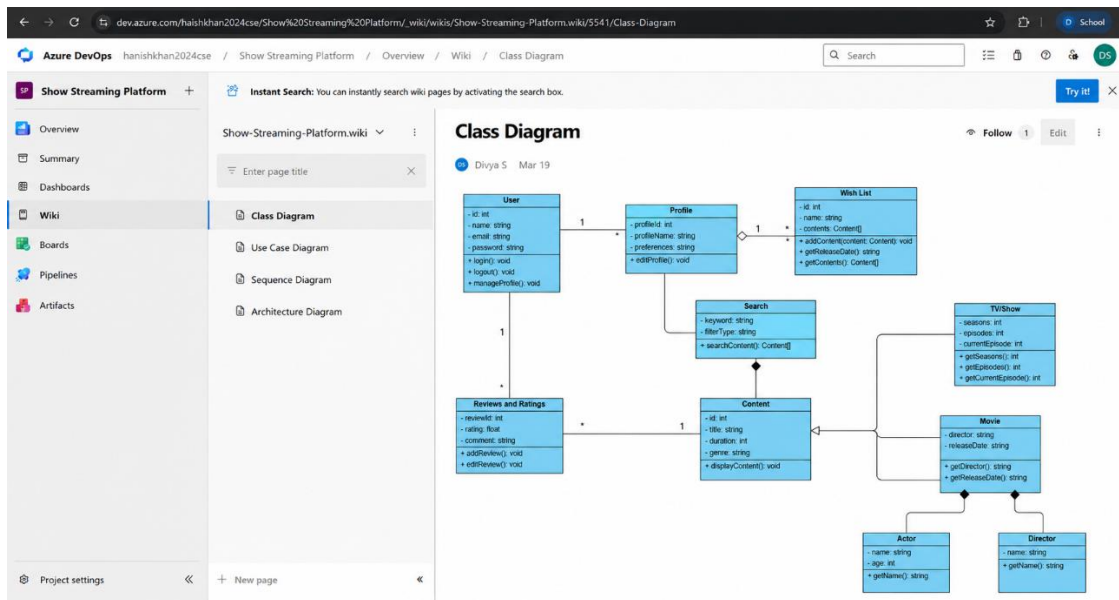
EXP NO: 7

CLASS DIAGRAM

AIM:

To Design a Class Diagram for the given Project.

7. CLASS DIAGRAM



RESULT:

The Class Diagram is designed Successfully for the Shows Streaming Platform.

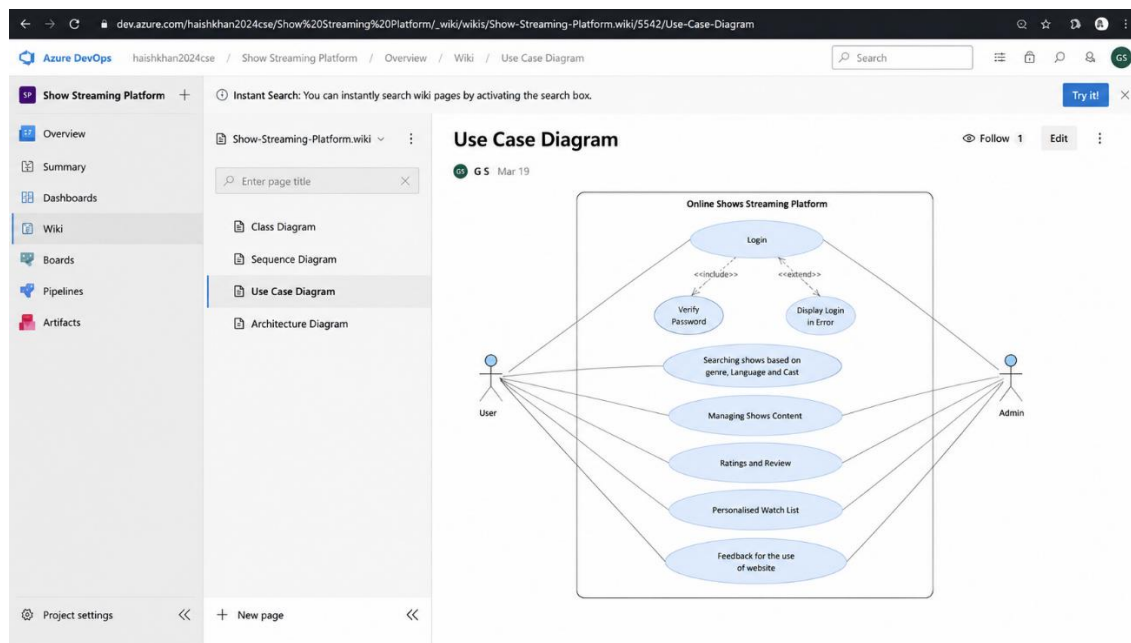
EXP NO:8

USE CASE DIAGRAM

AIM:

To design a Use case Diagram for the given Project.

8. USE CASE DIAGRAM :



RESULT:

The Use case diagram is designed Successfully for the Shows Streaming Platform.

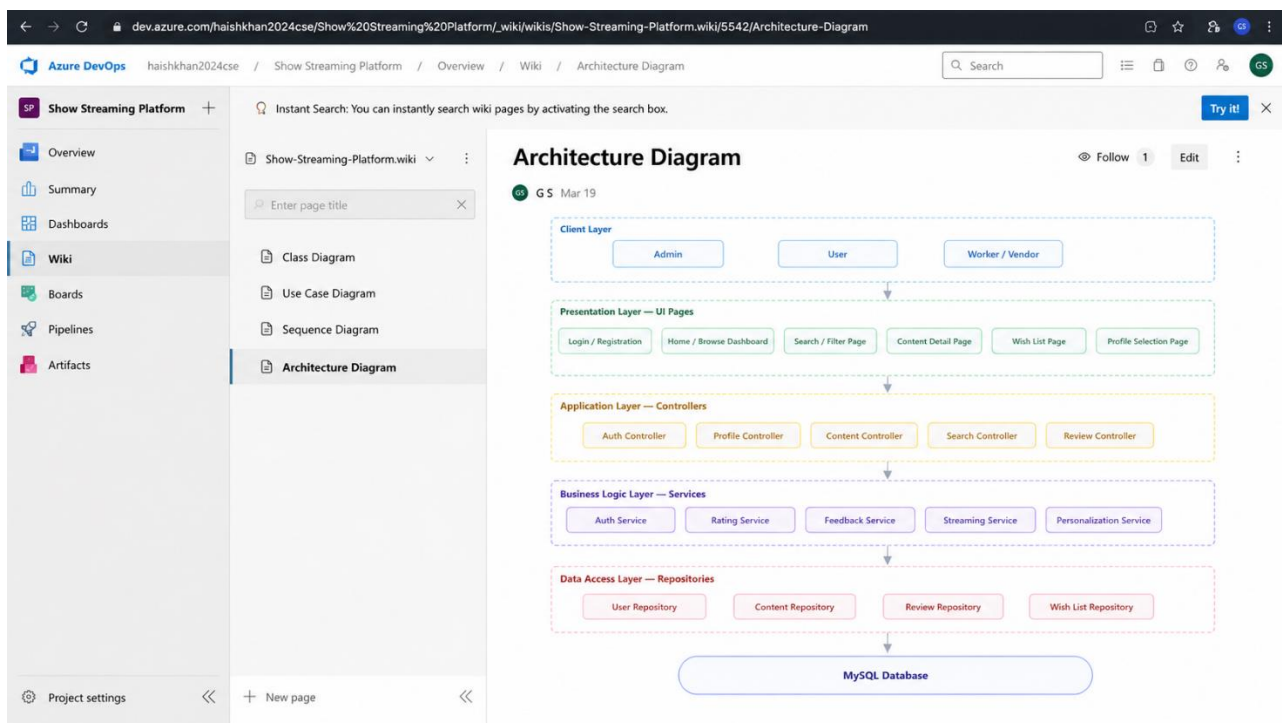
EXP NO:9

ARCHITECTURE DIAGRAM

AIM:

To design a Use case Diagram for the given Project.

9. ARCHITECTURE DIAGRAM :



RESULT:

The Architecture diagram is designed Successfully for the Shows Streaming Platform.

EXP NO: 10	TESTING AND DEPLOYMENT
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AIM:

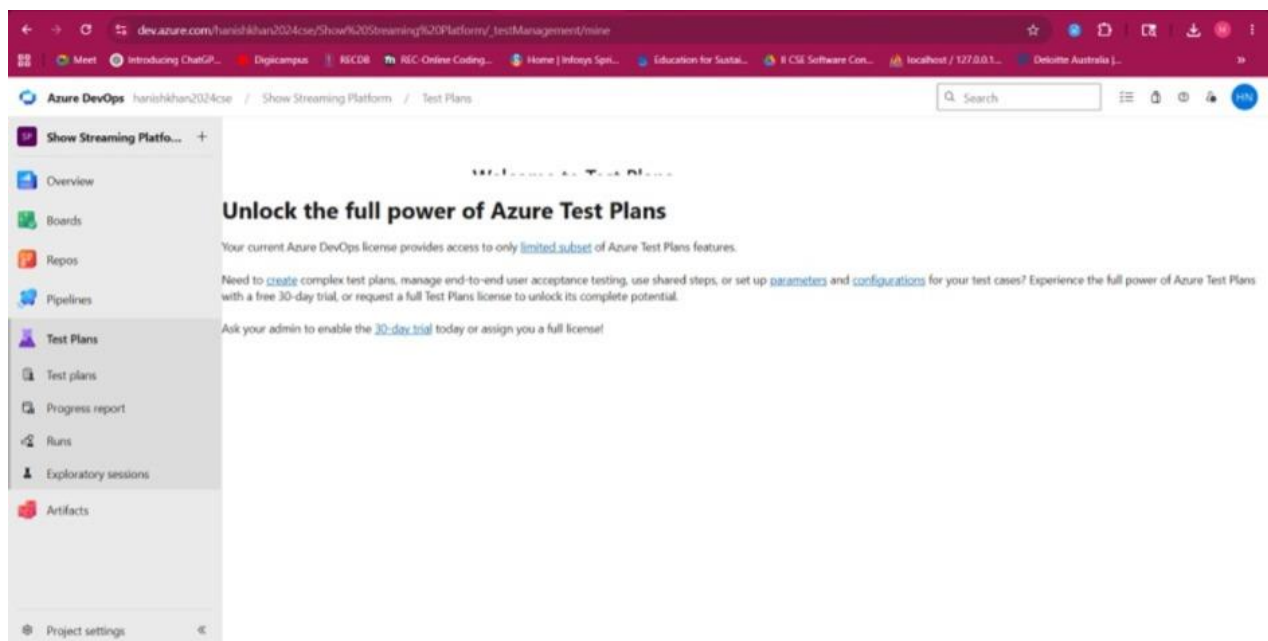
To perform testing and deployment activities for the Show Streaming Platform project.

THEORY :

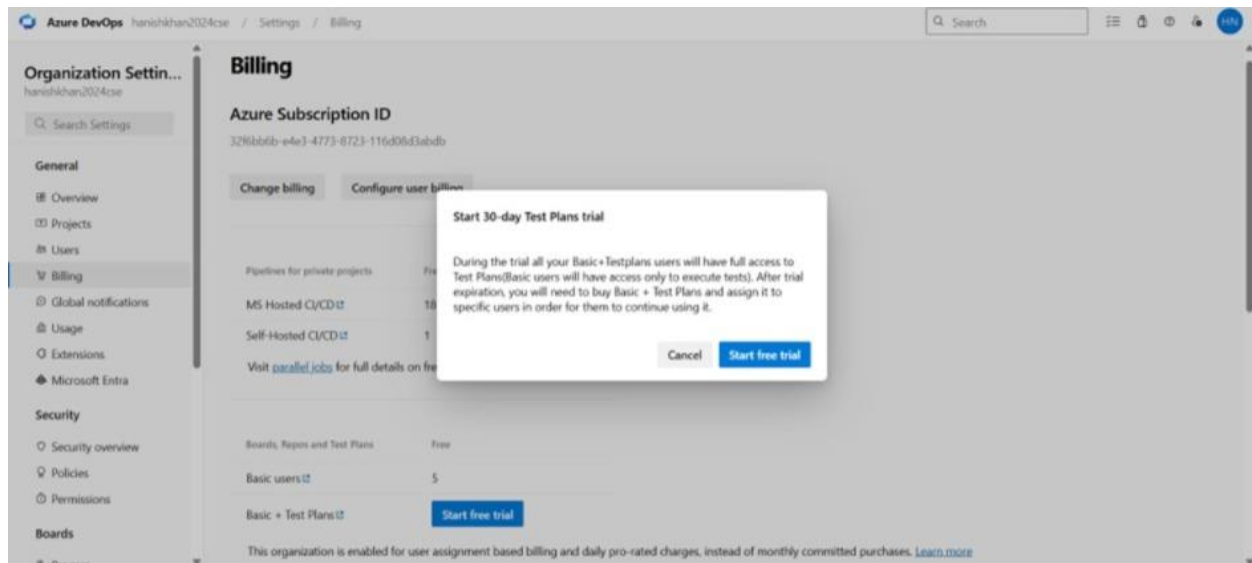
Testing is an important phase in software development used to verify that the application functions correctly and satisfies user requirements. Deployment is the process of making the developed application available for execution in a configured environment.

Azure DevOps provides integrated services for project management, testing, version control, continuous integration, and deployment. It helps teams manage the software development lifecycle efficiently through automated workflows and collaboration tools.

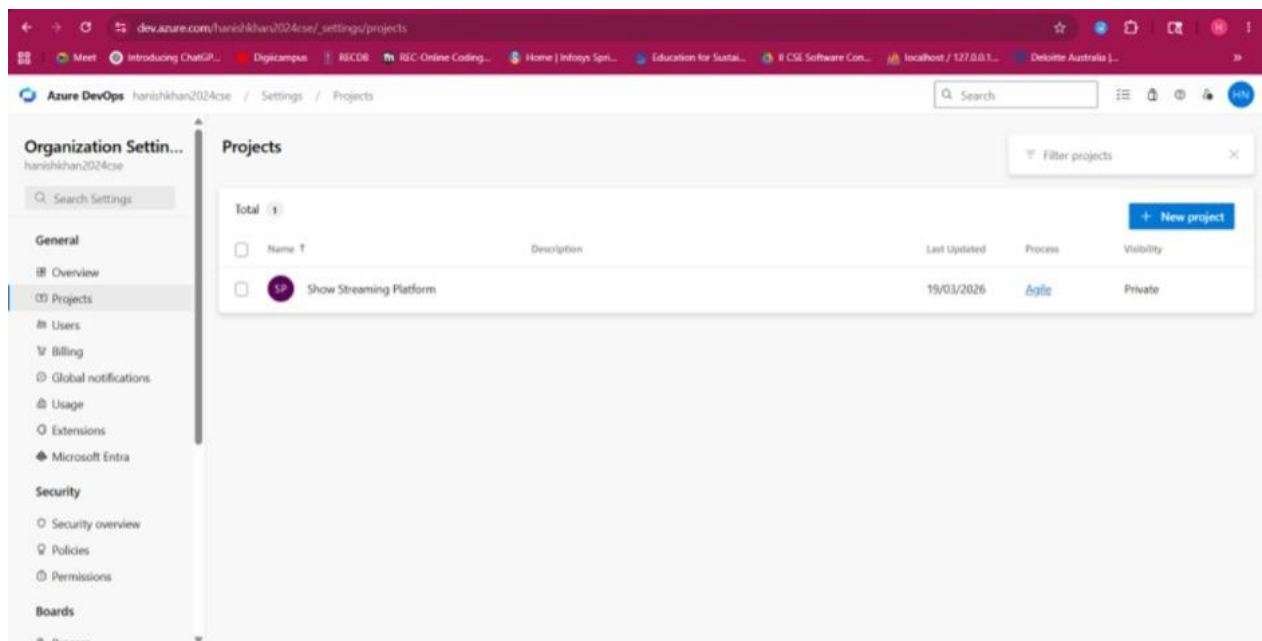
TESTING ACTIVITIES PERFORMED :



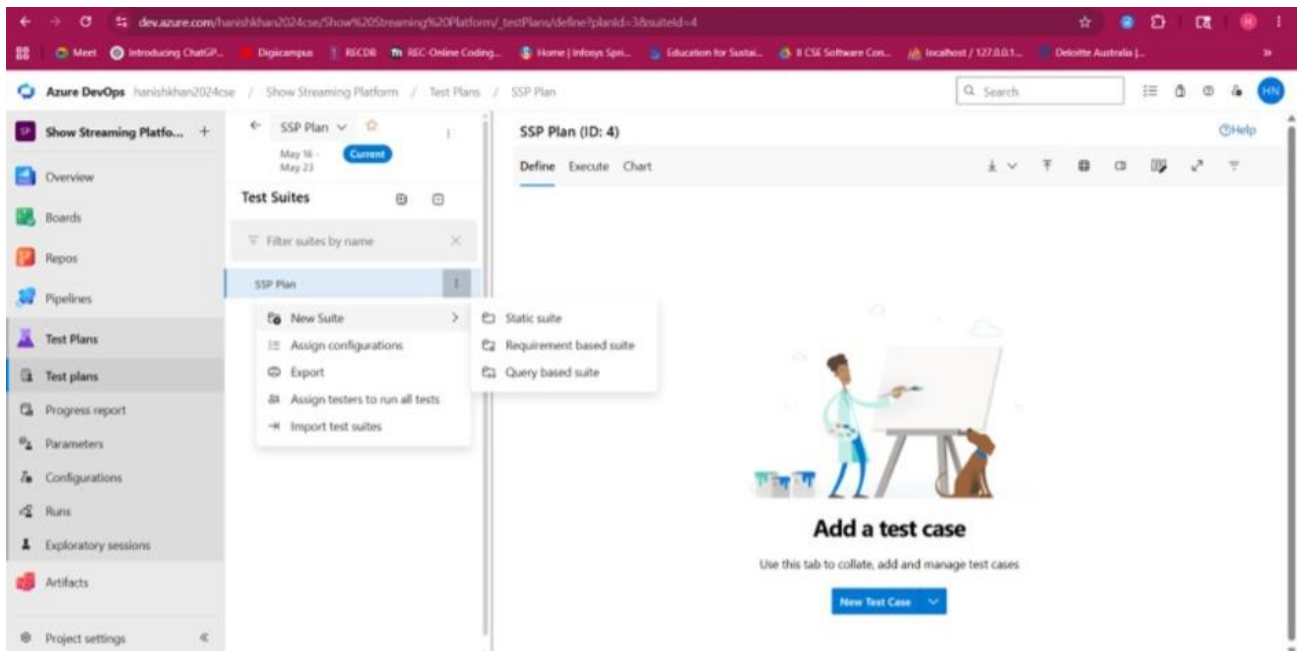
The functional gateway confirmation screen inside the Azure DevOps Test Plans hub.



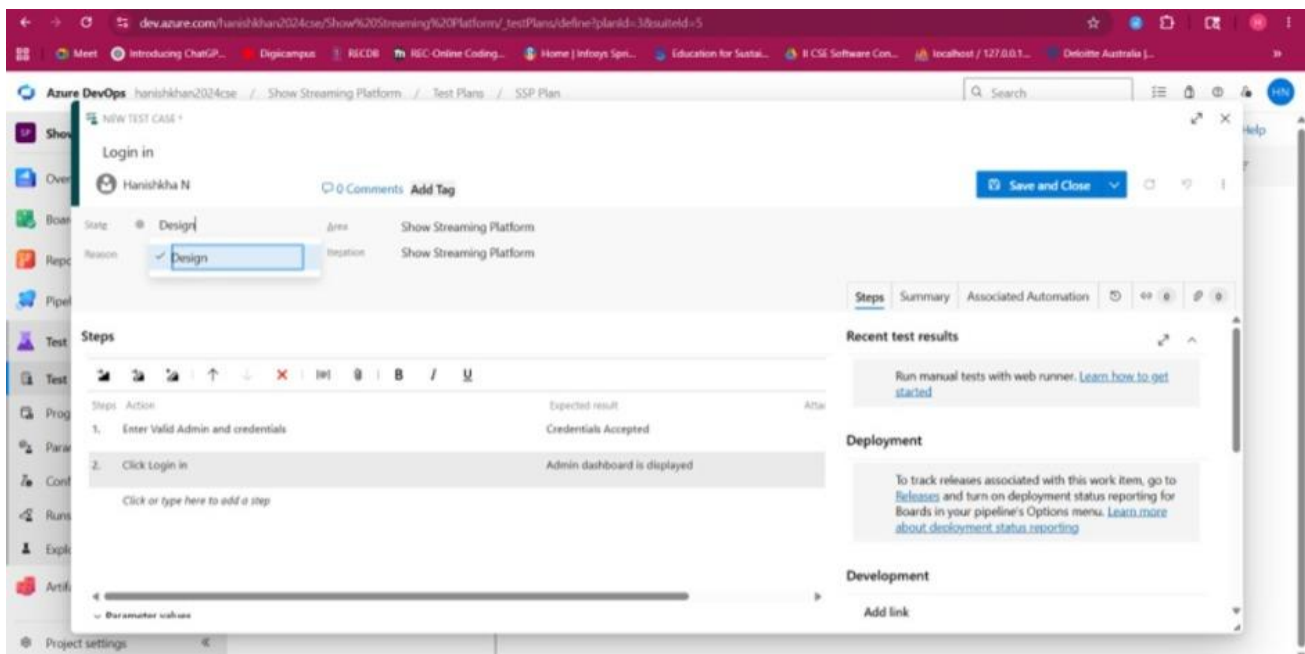
This step involves managing and securing the necessary licensing environment to enable full testing management capabilities.



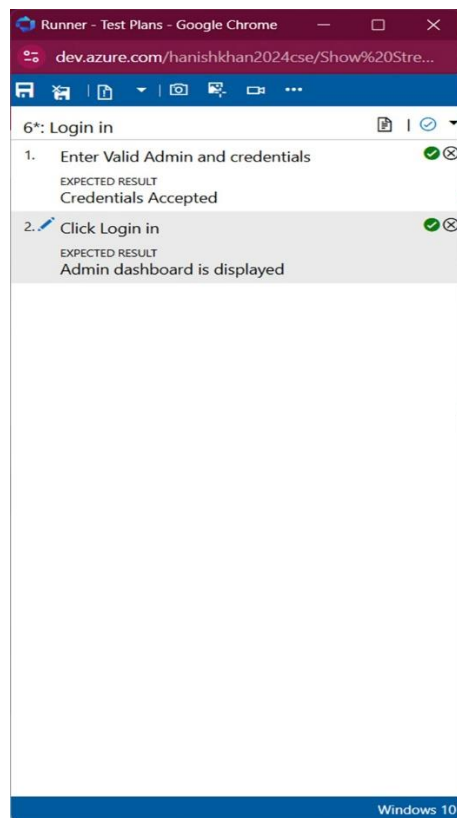
Represents the initial project verification workspace setup inside Organization Settings.



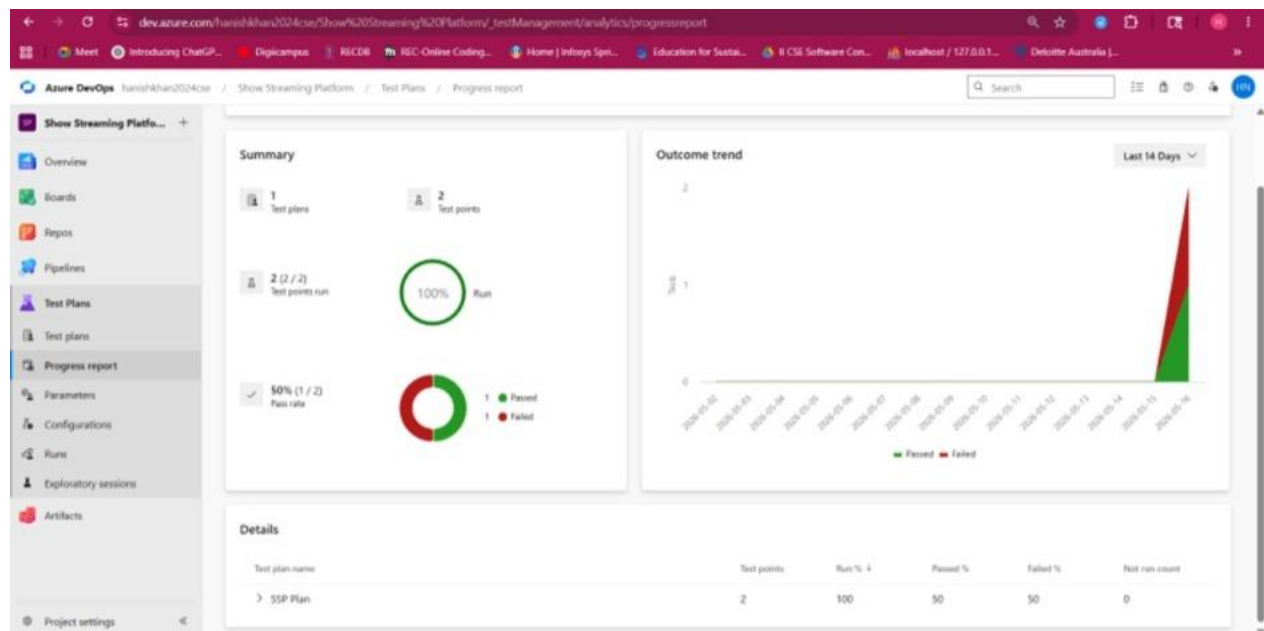
Involves initiating the testing lifecycle within Azure DevOps by creating a structured test suite under the main test plan.



The Organization Settings dashboard



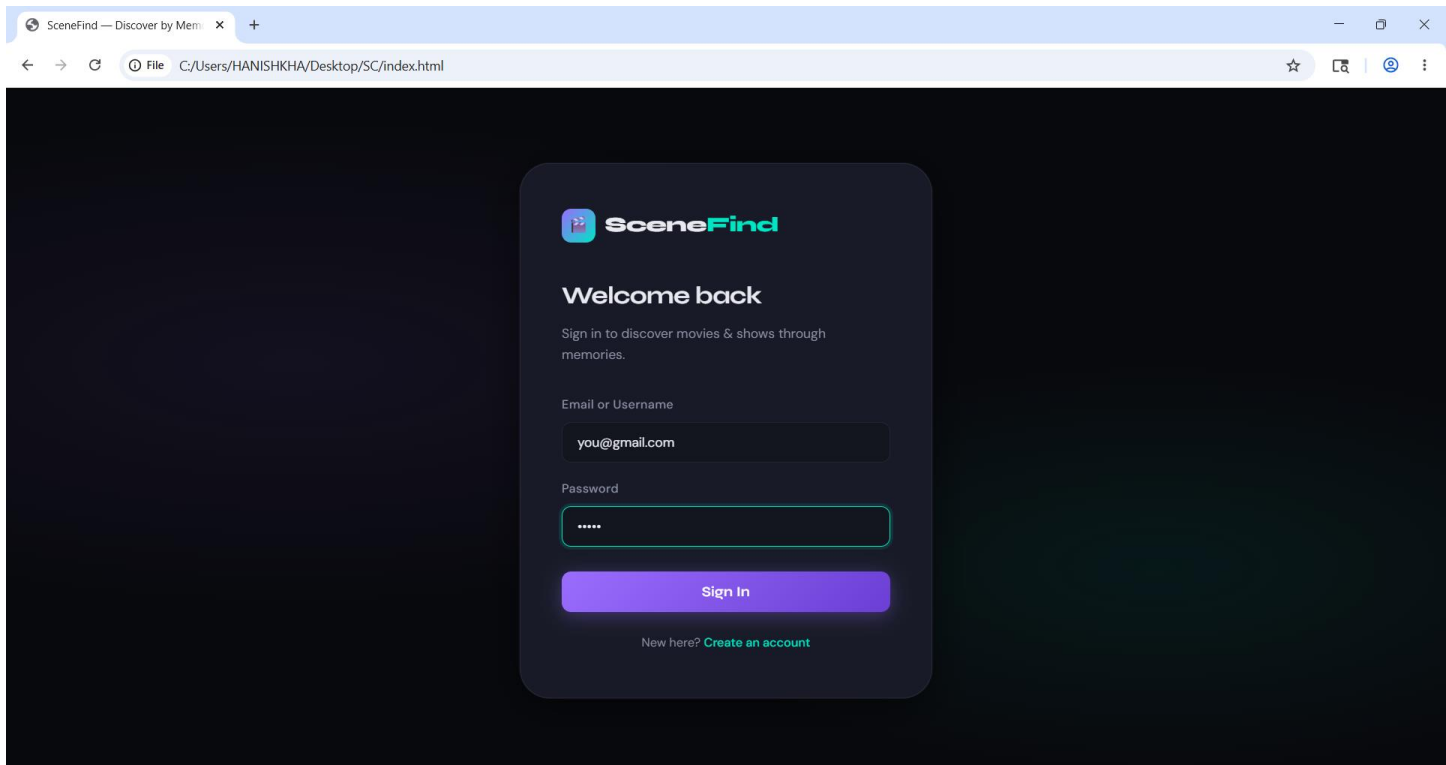
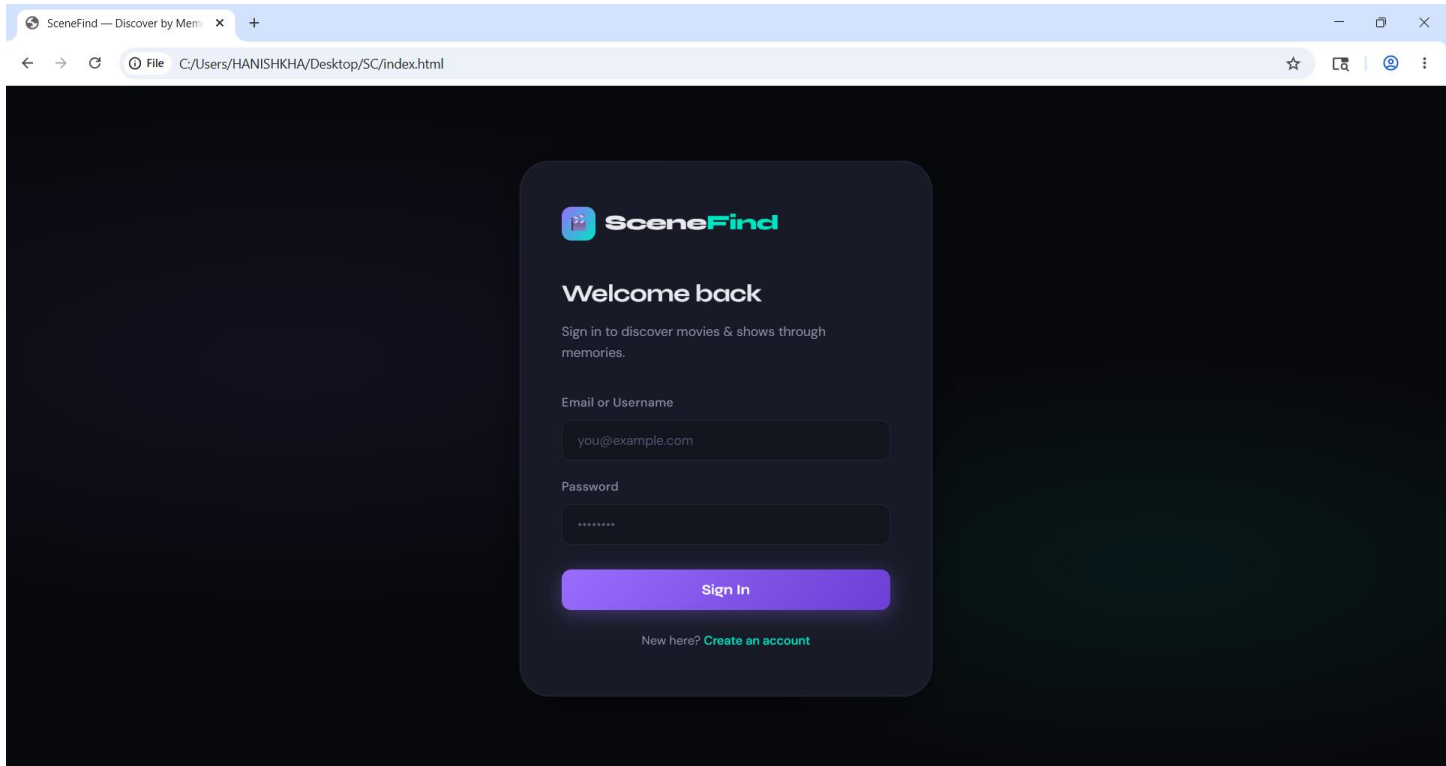
The standalone Azure DevOps Test Runner window actively executing a test case titled "Login in"

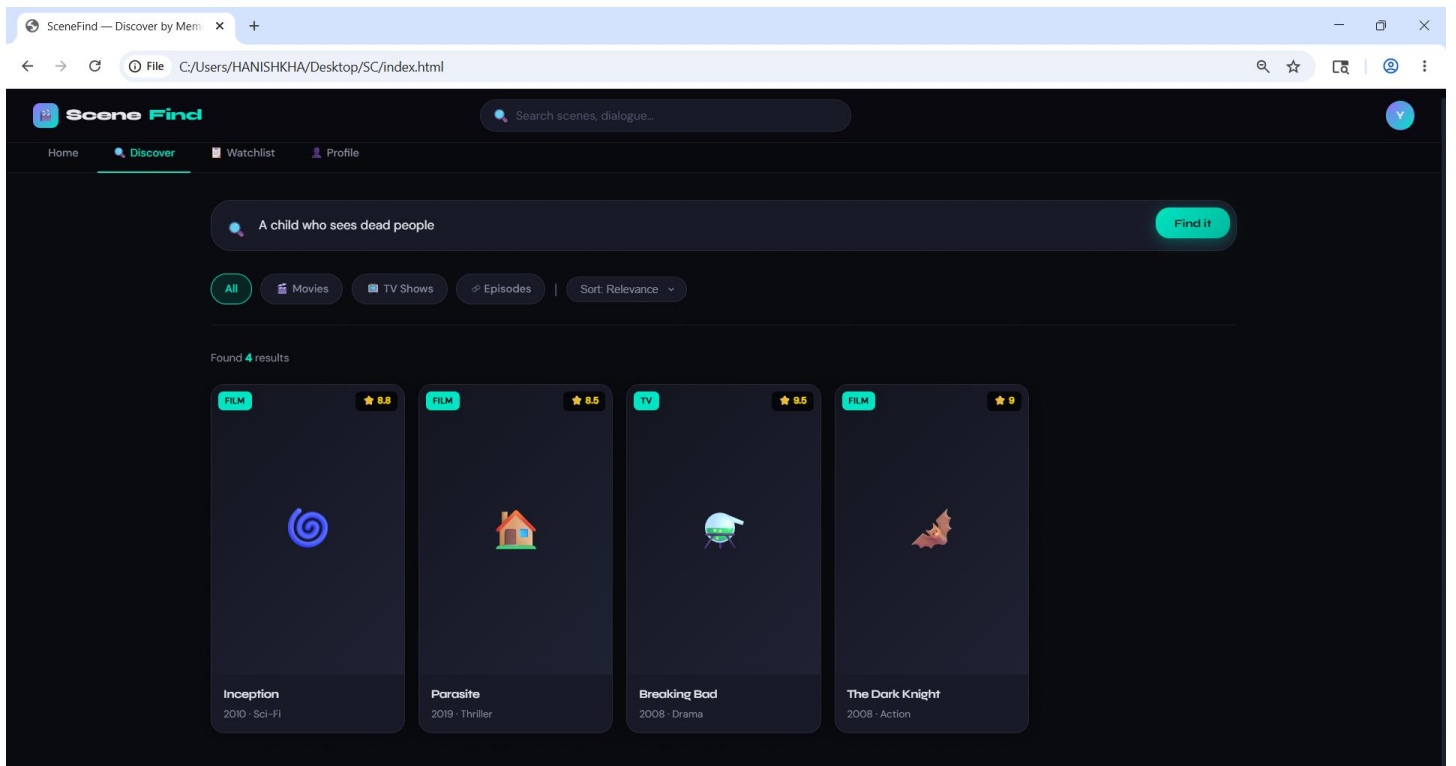
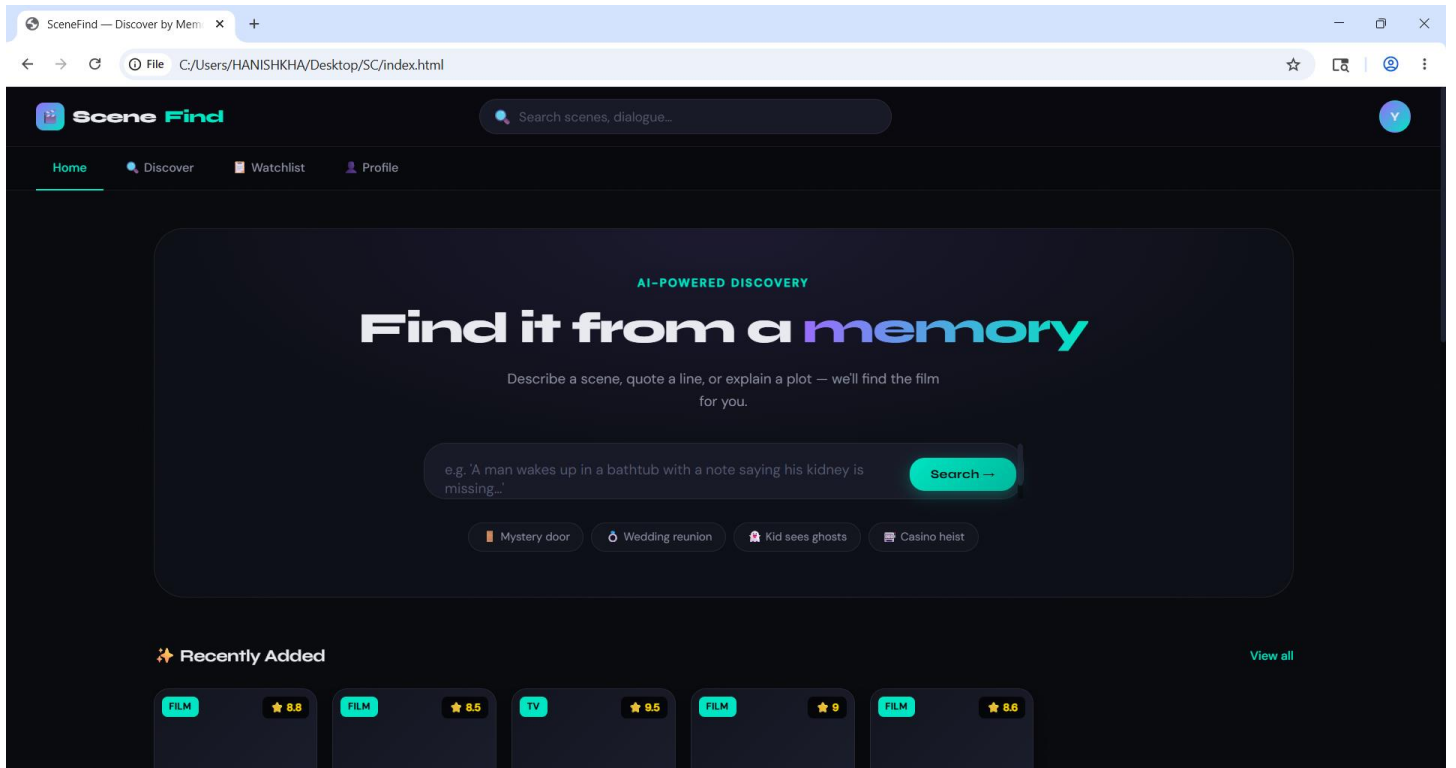


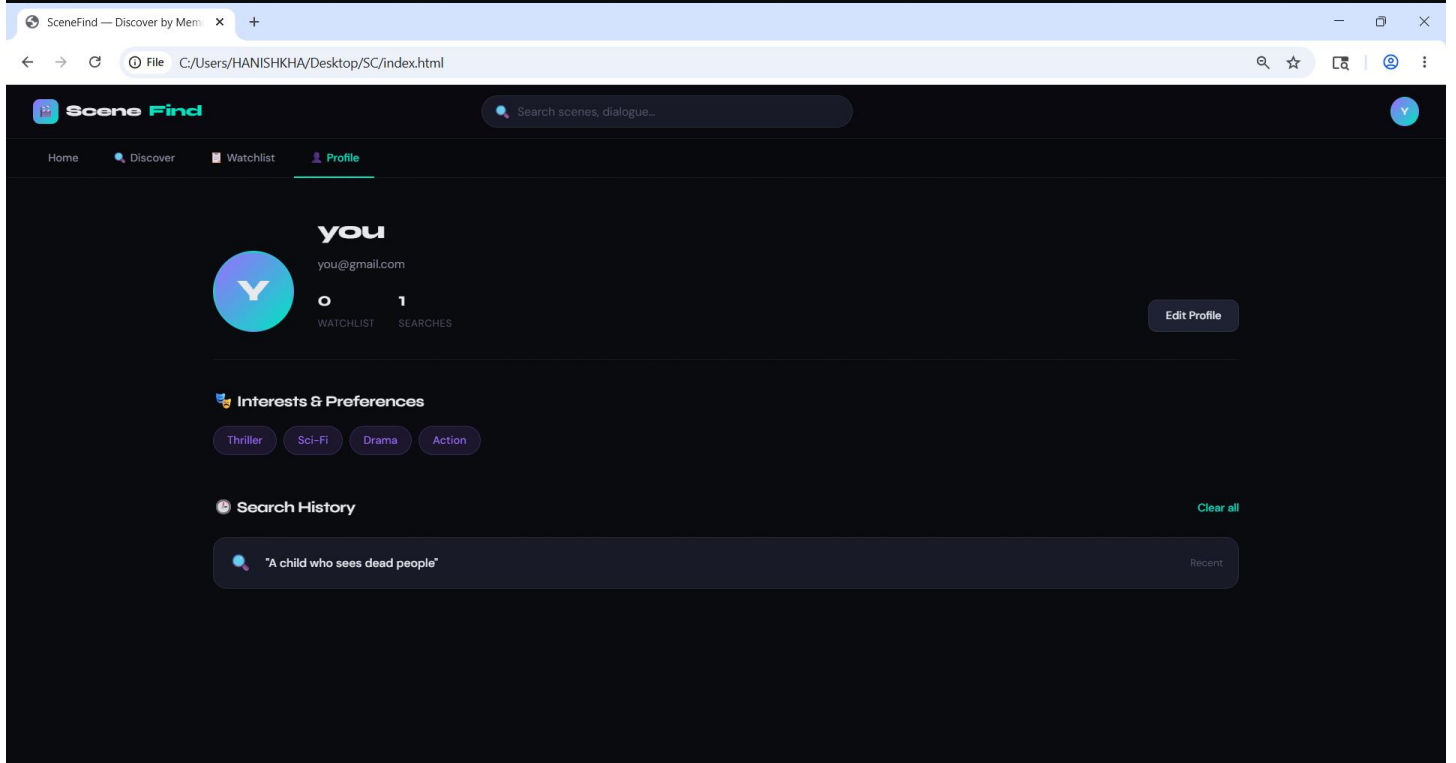
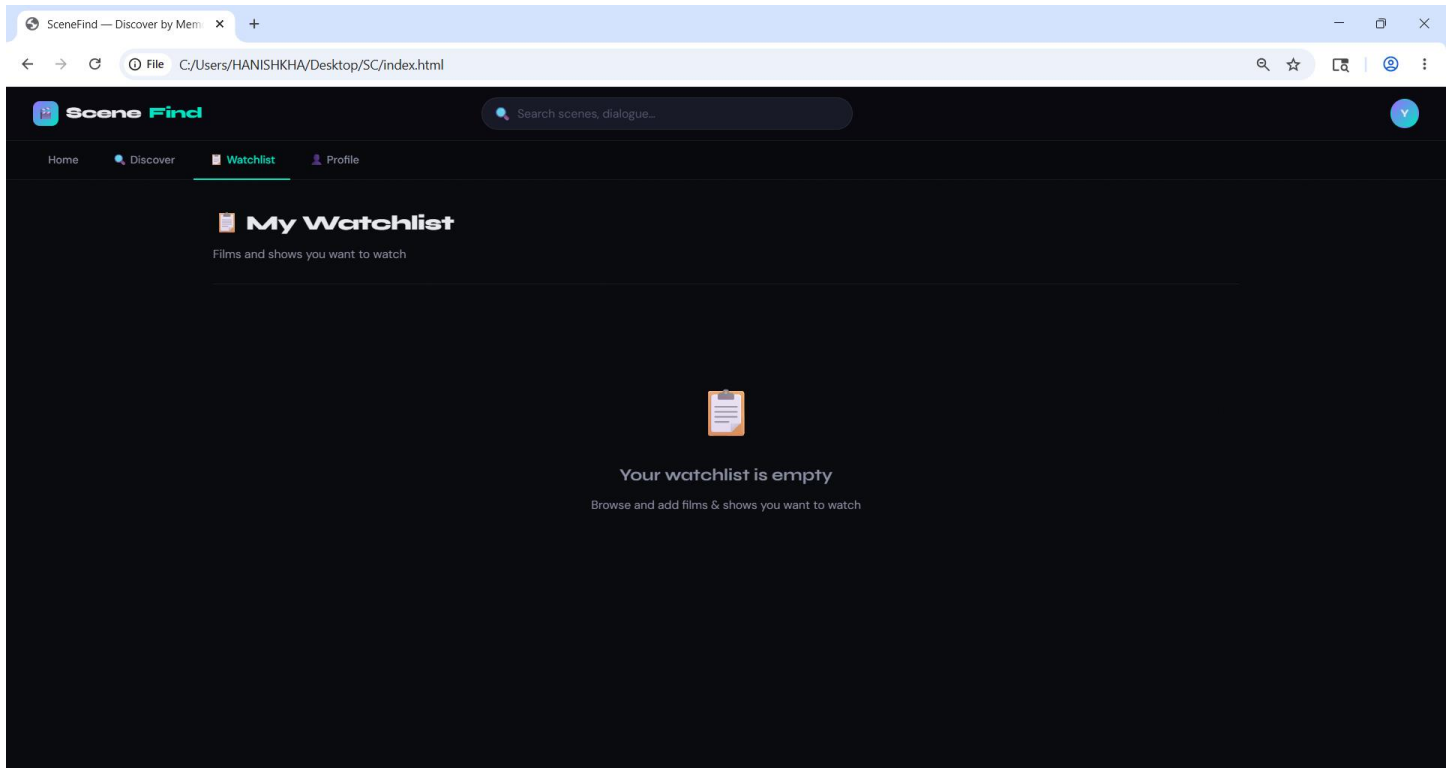
Displays the Test Plans Progress Report analytics dashboard for the streaming platform

DEPLOYMENT ACTIVITIES PERFORMED:

- Uploaded project files to Azure DevOps repository
- Managed project tasks and user stories using Azure Boards
- Maintained project documentation and diagrams in Azure Wiki
- Configured project environment and database connectivity
- Verified successful application deployment and execution







ShowScope

HomeBrowseSearch

episodes

S1 · E1Pilot58 min

Diagnosed with terminal cancer, chemistry teacher Walter White teams with a former student to manufacture and sell crystal meth.

S1 · E2Cat's in the Bag...48 min

Walt and Jesse attempt to tie up loose ends after their first cook goes sideways, leaving them to dispose of a body and a still-living captive.

S1 · E3...And the Bag's in the River48 min

Walt fights with Jesse over his drug use, causing him to leave Walt alone with his prisoner. Walt must make a fateful decision.

Reviews

Your rating: 8/10

Share your thoughts (optional)

Post review

ShowScope

HomeBrowseSearch

Browse

Explore the full catalog by type and genre.

AllShowsMovies

AllActionAdventureAnimationComedyCrimeDramaFantasyHistoryHorrorMysterySci-FiSupernaturalThriller

SHOW9.0

Arcane

2021 · Animation

MOVIE7.6

Avatar: The Way of Water

2022 · Sci-Fi

SHOW8.9

Better Call Saul

2015 · Crime

SHOW9.5

Breaking Bad

2008 · Crime

SHOW8.8

Dark

2017 · Sci-Fi

MOVIE8.6

Dune: Part Two

2024 · Sci-Fi

SHOW9.2

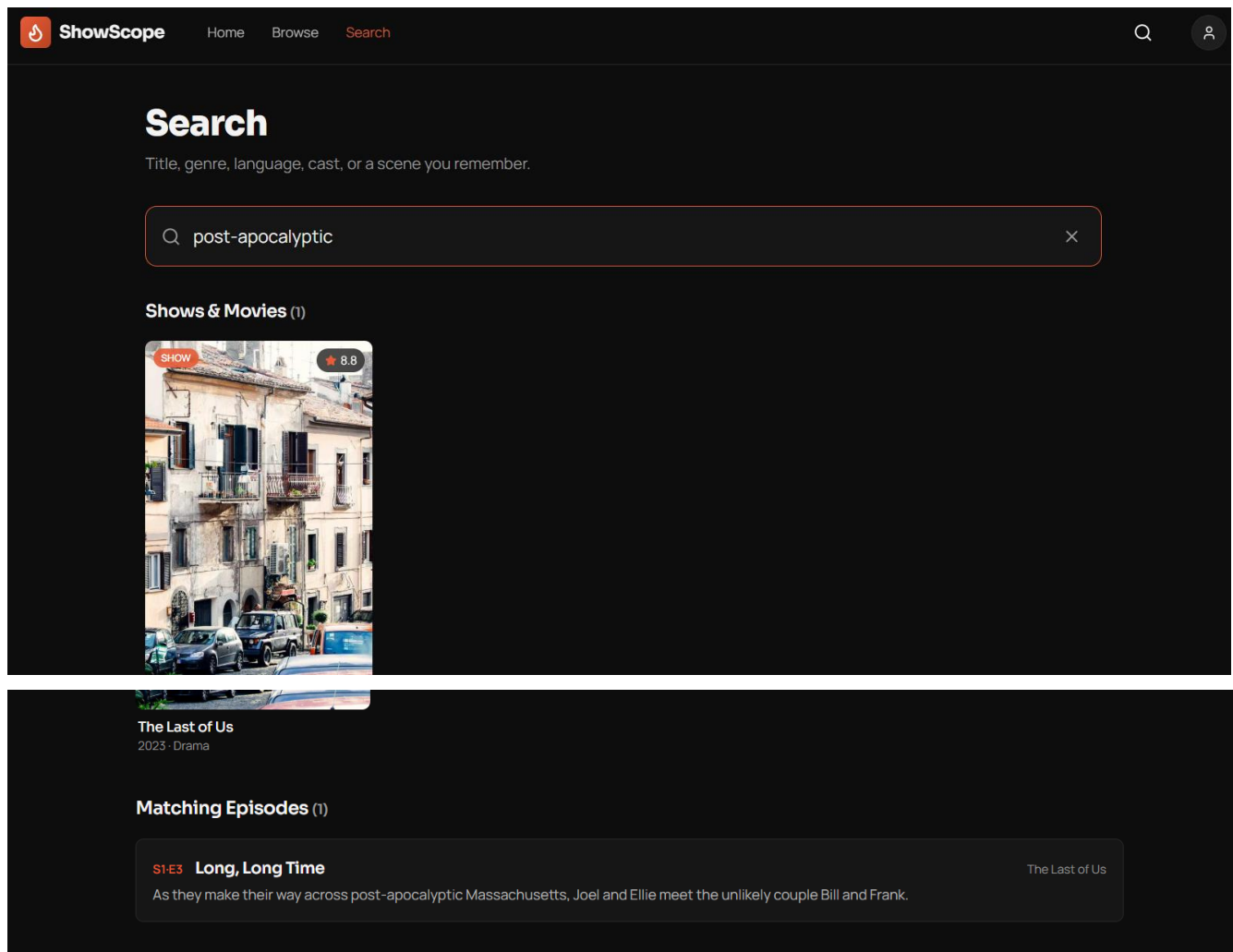
SHOW8.4

MOVIE8.8

MOVIE8.7

SHOW8.3

MOVIE8.4



RESULT :

Successfully tested and deployed the Show Streaming Platform project using Azure DevOps with all major functionalities working properly.